

Device Architectures for Low Voltage and Ultrafast Graphene Integrated Phase Modulators

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Abstract—The atomic layer thin geometry and semi-metallic band diagram of graphene can be utilized for significantly improving the performance matrix of integrated photonic devices. Its semiconductor-like behavior of Fermi-level tunability allows graphene to serve as an active layer for electro-optic modulation. As a low loss metal layer, graphene can be placed much closer to active layer for low voltage operation. In this work, we investigate hybrid device architectures utilizing semiconductor and metallic properties of the graphene for ultrafast and energy-efficient electro-optic phase modulators on semiconductor and dielectric platforms. (1) Directly contacted graphene-silicon heterojunctions. Without the oxide layer, the carrier density of graphene can be modulated by direct contact to silicon layer, while silicon intrinsic region stays mostly depleted. With doped silicon as electrodes, carriers can be quickly injected and depleted from the active region in graphene. The ultrafast carrier transit time and small RC constant promise ultrafast modulation speed (3 dB bandwidth of 67 GHz) with an estimated $V_{\pi} \cdot L$ of 1.19 V·mm. (2) Graphene integrated lithium niobate modulator. As a transparent electrode, graphene can be placed close to integrated lithium niobate waveguide for improving coupling coefficient between optical mode profile and electric field with minimal additional loss (4.6 dB/cm). Numerical simulation indicates a $2.5 \times$ improvement of electro-optic field overlap coefficient, with an estimated V_{π} of 0.2 V.

Index Terms—Graphene, silicon photonics, Lithium Niobate, p - n junction, phase modulator, Mach–Zehnder interferometer.

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I. INTRODUCTION

AFTER the invention of on-off key electro-optic modulation, phase modulator is then developed for converting electrical signals onto optical domain for long haul fiber communication. In modern telecommunication networks, advanced modulation formats built on the combination of amplitude and phase modulators can manifest the information throughput [1]–[5]. High-speed integrated optical phase modulators have been implemented on different material platforms of silicon (Si) [6], lithium niobate (LN), III-V materials and electro-optic polymers. The strong electro-optic response enables low drive voltage for III-V semiconductor and polymer-based modulators. However, their scalability, robustness and CMOS compatibility set limitations on their application. Wafer-scale Si and LN thin film-on-insulator structures are commercially available for integrated photonic devices and system implementation. However, their smaller electro-optic coefficient needs to be compensated by special device designs for achieving CMOS drivable voltage and sub-Terahertz bandwidth for meeting today's tele-comm and data-comm demands.

Atomic thin graphene (G), with zero bandgap, atomic layer thickness, flexibility, and mechanically strong in-plane structure, can be easily integrated onto those monolithic material platforms for improving the device performance matrix. G has been demonstrated as an active media for both amplitude [11]–[15] and phase modulators [16], [17] with metal electrodes. The modulation bandwidth is measured to be between sub GHz to 30 GHz, which is well below the material limitation of ~ 500 GHz bandwidth (2.2 ps response time) in G [18]. The bandwidth limitation is originated by either RC constant or carrier transient time. Those electronic limitations by device design are well studied in G photodetectors with G-metal contact [19]–[23] and G-Si Schottky junction [24]–[26]. In previous works, we have demonstrated that directly contacted G-Si p - i - n junctions can utilize both ultrafast carrier transportation in G and small RC constant in p - i - n junction for near 100 GHz operation bandwidth [8]–[10]. In this work, we will explore the device architecture for high-speed modulators. Distinguished from other works on G-Si integrated optoelectronics, our device leverages the pristine surface provided by foundry, which allows efficient carrier injection and depletion on directly contacted G-Si heterojunction [16], [17]. By optimizing doping profiles in Si and graphene contact

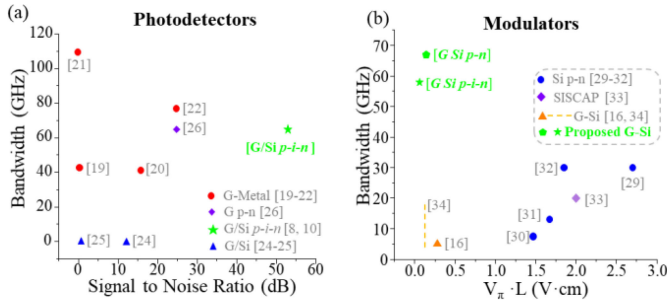


Fig. 1. Performance matrix comparison of (a) G-Si Photodetectors. (b) Si and G-Si modulators [8], [10], [16], [19]–[22], [24]–[26], [29]–[34].

region, photonic crystal (PhC) waveguides based Mach-Zehnder modulator (MZM) has numerically shown with a very low $V_{\pi} \cdot L$ (1.19V·mm) and fast EO response (67 GHz) for high speed and efficient modulator applications.

Unlike semiconductor-based phase modulators, the modulation bandwidth in lithium niobate is not restricted by the carrier transient time. Many low drive voltage LN modulators are experimentally demonstrated [4], [27], [28] with metal electrodes. However, the figure-of-merit $V_{\pi} \cdot L$ is still fundamentally limited by the trade-off between electrical and optical designs: closer electrodes lead to lower electro-optic tuning efficiency but higher optical loss through metal scattering and absorption. Also, conventional lithography and dry-etching methods can contaminate the etching chamber and need to be finely optimized the recipe for achieving a smooth sidewall. In this section, we develop a hybrid modulator design with graphene as an optically transparent electrode layer for maximizing the overlap between electric field and optical field, based on an optical design with maximized confinement factor in LN.

II. DIRECTLY CONTACTED GRAPHENE-SILICON PHOTONIC CRYSTAL PHASE MODULATORS

Si photonic platform offers a high yield, high density, and low loss passive components for large-scale integrated photonic circuits, and seamless compatibility with CMOS electronic systems. Developing a back-end-of-line compatible process for G, while maintaining their superior performance matrix, is critical for low-cost, large-scale active Si photonic devices. In the directly contacted G-Si hybrid device, the carrier transportation will be dominated by high mobility of G, and the RC constant limit becomes a limiting factor for the device speed [10].

Here we compare the performance matrix of proposed device architecture for both photodetector and modulator applications (Fig. 1). For photodetectors [19], [20], G-metal heterojunction photodetectors have been demonstrated which can achieve over 100 GHz but with very low Signal-to-Noise Ratio (SNR) less than 1 dB [21], or attain 25 dB SNR and over 70 GHz bandwidth [22]. Other works using graphene silicon directly contacted photodetectors have SNR less than 15 dB and very low bandwidth [24], [25]. G integrated on a slot waveguide to create a *p-n* junction is demonstrated with 65 GHz bandwidth, but SNR is still not very high [26]. Comparing with other types of graphene-based photodetector, our work of G directly contacted

Si *p-i-n* junction based on G absorption not only offers very high SNR over 50 dB but provides bandwidth over 60 GHz as well [8]. For modulators based on monolithic silicon platform, such as Si-insulator-Si capacitor, or traveling wave Mach-Zehnder interferometer lateral *p-n* junction, they cannot achieve high bandwidth while at a very low $V_{\pi} \cdot L$ [29]–[33]. Those are restricted by mobility in silicon and limited refractive index change. For the graphene silicon modulators [16], [34], most are carrier accumulation-based modulation mechanism which may achieve high efficiency but limited by speed because this suffers from high junction capacitance and slow free-carrier recombination [35]. Simulation and numerical results of our devices which base on directly contact G-Si *p-i-n/p-n* configuration show high bandwidth and low $V_{\pi} \cdot L$ which has great potentials for high speed and efficiency applications.

A. Graphene-Silicon *p-n* Modulators

Modulation in silicon is realized by plasma dispersion effect to induce the refractive index change [1], [35], [36]. There are three types of mechanisms in a silicon waveguide to achieve carrier density changes: carrier injection in a forward-biased *p-i-n* junction structure, carrier accumulation in a MOS capacitor structure, and carrier depletion in a reverse-biased *p-n* junction structure [1], [35]. Carrier injection is the most efficient modulation mechanism, but it is hard to achieve high speed because it suffers from high junction capacitance and slow free-carrier recombination. Carrier accumulation in MOS capacitors can achieve high modulation efficiency, but the speed of MOS-type modulator is limited by the high capacitances from a thin oxide layer. Carrier depletion has very low modulation efficiency, but can achieve high-speed performance by optimizing doping profiles to reduce the junction capacitance [35]. Currently, most efforts for studying G-Si modulators use the oxide structure of charge accumulation scheme to enhance the efficiency while has limited speed [16].

As an example, a Mach-Zehnder modulator (MZM) with two arms of proposed G-Si phase modulator is illustrated in Fig. 2(a) [8], [37], [38]. Photonic crystal (PhC) waveguides have a lattice constant of 440 nm, different radii of 135 nm and 140 nm [39]. Each arm of MZM can be driven by electrical signals to achieve refractive index change under bias. Simulated optical mode profile of TE mode at 1550 nm can be seen in Fig. 2(b). Here, to explore the high speed and high efficient performance based on silicon modulator, two doping profiles are proposed for graphene silicon hybrid modulators: G-Si *p-n* junction under reverse bias and G-Si *p-i-n* junction under forward bias (Figs. 2(c) and 2(d)). A Wider N region (4 μm) region makes the optical mode profile overlap with the strong built-in electric field around G and Si/n region to enhance the electric-optical interaction. Single layer graphene will be transferred onto PhC WG covering P+, P and N region to achieve that Figs 2(b) and 2(c) [10]. Also, for *p-i-n* junction, graphene covers from P+ to intrinsic region (width of 6 μm) good electric-optical interaction. For the doping profile, lightly doped P ($1 \times 10^{18} \text{ cm}^{-3}$) and N ($1 \times 10^{18} \text{ cm}^{-3}$) region in the middle to form the *p-n* junction (Fig. 2(c)). High doped region of P+ ($5 \times 10^{18} \text{ cm}^{-3}$) and N+ ($5 \times 10^{18} \text{ cm}^{-3}$) are at each

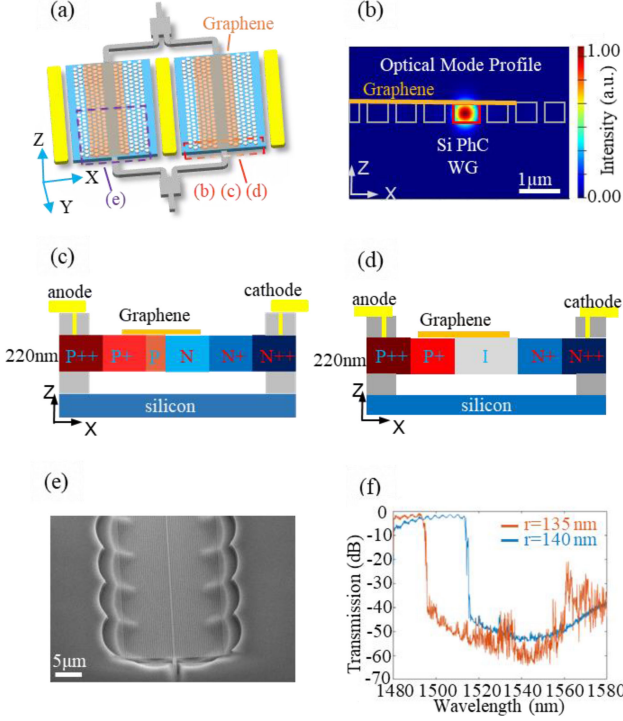


Fig. 2. Device schematics. (a) Schematic of proposed G-Si photonic crystal waveguide (PhC WG) MZM modulators. (b) Cross-sectional optical mode profile (TE mode) of PhC WG at 1550 nm. (c) Doping profile of graphene silicon p - n junction modulator. (d) Doping profile of graphene silicon p - i - n junction modulator. (e) SEM imaging of the foundry processed silicon PhC WG. (f) Measured transmission spectra of PhC waveguides after undercut. r : radii of the holes in the PhC slab [39].

side of P and N region to connect heavily doped P++ ($8 \times 10^{19} \text{ cm}^{-3}$) and N++ ($8 \times 10^{19} \text{ cm}^{-3}$) regions which form the ohmic contact with the anode and cathode, respectively. Intrinsic region has a lightly p doped concentration of 10^{16} cm^{-3} . SEM imaging showing the fabricated PhC WG is presented in Fig. 2(e). The measured transmission spectra of PhC waveguides with different radii can be seen in Fig. 2(f).

For this hybrid structure, effective index variation induced by the charge carrier density changes in both graphene and silicon under bias is numerically investigated. Lumerical DEVICE is used to simulate the charge carrier dynamics in the cross-section (Figs 2(c) and 2(d)) for this hybrid junction under bias. Charge carrier concentrations are exported from Lumerical DEVICE suite for numerical calculation and analysis. After transferred on the silicon substrate, graphene is electrostatically doped. Lateral bias across the p - n junction shifts G Fermi level and carrier density. G's conductivity, permittivity, and complex refractive index are numerically calculated by equations (1–3) [3], [40], and illustrated in Fig. 3(a).

$$\mu_c(n_s) = \text{sgn}(n_s) \hbar v_F \times \sqrt{\pi |n_s|} \quad (1)$$

$$\sigma(\omega) = \frac{\sigma_0}{2} \left(\tanh\left(\frac{\hbar\omega + 2\mu_c}{4k_B T}\right) + \tanh\left(\frac{\hbar\omega - 2\mu_c}{4k_B T}\right) \right)$$

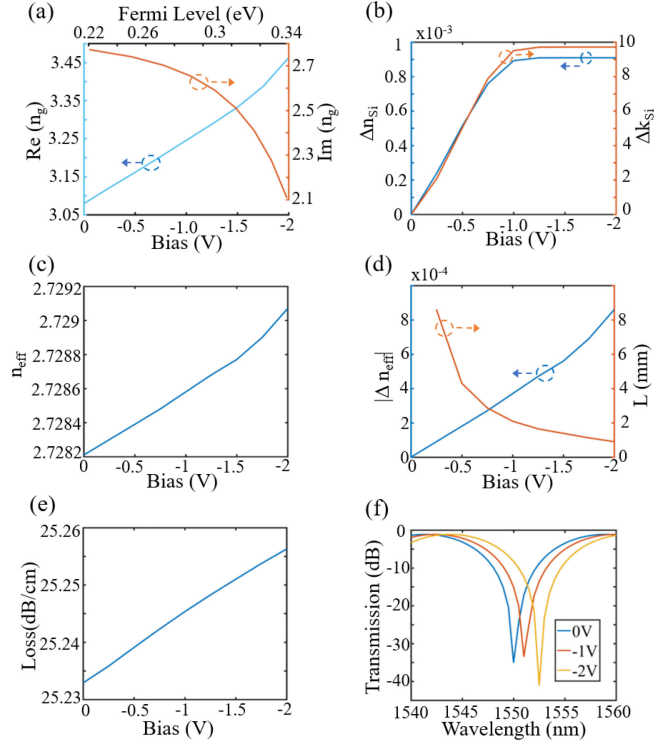


Fig. 3. G-Si p - n junction modulator. (a) G Fermi level and refractive index variation under reverse bias of 0 to -2V . (b) Real and imaginary part of silicon refractive index change under reverse bias from 0 to -2V . (c) Effective index variation of the hybrid structure under reverse bias from 0 to -2V . (d) Total effective index change and the length to achieve π phase shift under reserve bias from 0 to -2V . (e) Loss of the hybrid waveguide with reserve bias from 0 to -2V . (f) Transmission spectrum of driving one arm of MZM around 1550 nm under bias 0 to -2V .

$$-i \frac{\sigma_0}{2\pi} \ln \left[\frac{(\hbar\omega + 2\mu_c)^2}{(\hbar\omega - 2\mu_c)^2 - 2(k_B T)^2} \right] + i \frac{4\sigma_0}{\pi} \frac{\mu_c}{\hbar\omega + i\hbar\Gamma} \quad (2)$$

$$n_g = \sqrt{\epsilon_g} = \sqrt{1 + \frac{i\sigma}{\omega t_g \epsilon_0}} \quad (3)$$

Under reverse bias, the real part of G refractive index increases with bias while the imaginary part decreases. In Si, refractive index change is induced by the carrier densities change under bias following the equations (4,5) [6], [7]. The real and imaginary part of silicon refractive index increase with reverse bias (Fig. 3(b)).

$$\Delta n(1550\text{nm}) = -5.4 \times 10^{-22} \times \Delta N^{1.011} - 1.53 \times 10^{-18} \times \Delta P^{0.838} \quad (4)$$

$$\Delta \alpha(1550\text{nm}) = 8.88 \times 10^{-21} \times \Delta N^{1.167} + 5.84 \times 10^{-20} \times \Delta P^{1.109} \quad (5)$$

Parameter specifications in equations (1-5) can be seen in the Table I.

Numerical calculation estimated that silicon dominates and contributes most that is about 99.6% to the total effective index change comparing with graphene (0.29%) since the optical mode

TABLE I
TERMS FOR GRAPHENE AND SILICON MODULATOR

Symbol	QUANTITY	Values and unit ^a
μ_c	Fermi level of graphene	eV
n_s	carrier density in graphene	cm^{-2}
$\hbar$	reduced Planck constant	$6.58 \cdot 10^{-16} \text{ eV} \cdot \text{s}$
v_F	Fermi velocity	10^6 m/s
σ_0	single layer graphene universal conductivity	$60.8 \mu\text{S}$
k_B	Boltzmann constant	$1.380649 \cdot 10^{-23} \text{ J/K}$
T	temperature	300 K
ω	angular frequency	Rad/s
Γ	scattering time	10 fs
n_g	refractive index of graphene	
ε_g	permittivity of graphene	F
t_g	graphene thickness	1 nm
ε_0	vacuum permittivity	$8.85 \cdot 10^{-12} \text{ F/m}$
ΔP	hole concentration change in silicon	cm^{-3}
ΔN	electron concentration change in silicon	cm^{-3}
Δn	real part of refractive index change in silicon	
$\Delta\alpha$	absorption change in silicon	dB/cm

^a eV = electron volt, S = Simens; s = second, F = Farad.

profile is mostly confined in Si region (Fig. 3(b)). The change of total effective index of this hybrid structure is calculated up to 10^{-3} scale with reverse bias (Fig. 3(c)). The length required to achieve a π phase shift can be obtained in Fig. 3(d). The minimum $V_\pi \cdot L$ is $2.15\text{V} \cdot \text{mm}$ ($2.87\text{mm} \cdot 0.75\text{V}$). This is due to significant variation of carrier densities in directly contacted silicon and graphene under bias. The loss of this hybrid waveguide kept around 25.24 dB/cm at different bias (Fig. 3(e)). Assuming the length of each arm is $250 \mu\text{m}$, the transmission spectra of MZM are calculated with different reverse bias applied on one arm (Fig. 3(f)) [41]. About 20 dB and 30 dB extinction ratio can be achieved with reverse 1V and 2V bias respectively.

In a Si p - n junction, space charge region will be formed as the reverse bias is applied [1]. The width space charge region increases with the increasing built-in electric field induced by charge induced doping [8]. This is able to reduce the equivalent capacitance of the junction component, further reducing the RC constant and increasing the S_{11} bandwidth of the device [10]. Fig. 4(a) is the built-in electric field along z -direction (E_z) with lateral reverse bias. This reduces the junction capacitance and thus RC constant. The RC constant limited S_{11} response has been simulated using Advanced Design system (ADS 2012, by Keysight Technology) under different reverse bias [42]. In Fig. 4(b) as the reverse bias increases, the decreasing junction capacitance leads to the increase of S_{11} 3 dB bandwidth. For G-Si heterogeneous structure, the overall 3 dB bandwidth of electrical-optical (EO) response is determined by RC constant limited bandwidth and carrier transit time limited bandwidth

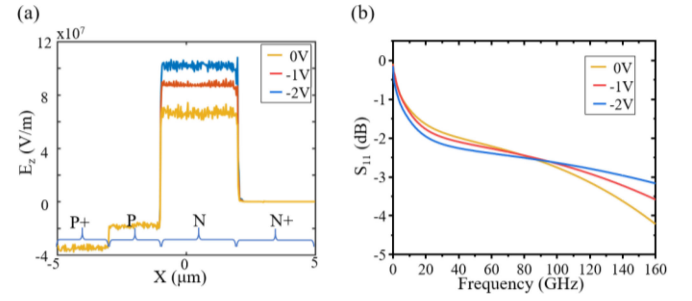


Fig. 4. (a) E_z component of built-in electric fields at graphene layer under reverse bias for G-Si p - n junction modulator. (b) Simulated S_{11} response G-Si p - n junction modulator under bias 0 to -2V.

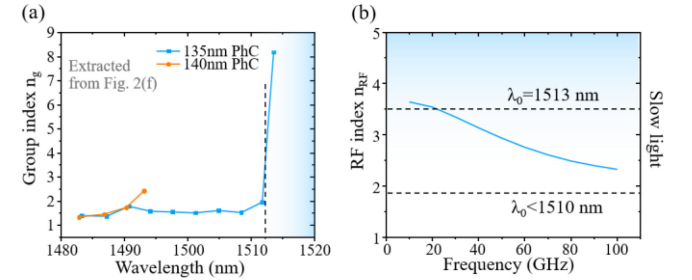


Fig. 5. (a) Extracted optical group index of PhC WG. (b) RF index spectra from 10 GHz to 100 GHz, compared to group index at the center wavelength in the slow light region (beyond 1520 nm).

(equation (6)) [43], [44]:

$$f_{3 \text{ dB}} \approx \frac{1}{\sqrt{\frac{1}{f_{RC}^2} + \frac{1}{f_{tr}^2}}} \quad (6)$$

Where f_{RC} is the RC constant limited bandwidth which can be obtained (about 105 GHz at 0V) from simulated S_{11} in Fig. 4(b). f_{tr} is the transit time limited bandwidth which is determined by: $f_{tr} = \frac{0.44}{\tau_{tr}}$, where transit time $\tau_{tr} = \frac{\text{depletion region width}}{\text{carrier velocity}} = \frac{5 \mu\text{m}}{10^6 \text{ m/s}} = 5\text{ps}$ (depletion width is estimated $5 \mu\text{m}$ here, carrier velocity is 10^6 m/s). So the overall 3 dB bandwidth of EO response is mainly limited by carrier transit time and is about 67 GHz.

For optical and RF index, severe mismatching will affect the 3 dB bandwidth of EO response at high frequency (e.g. 100 GHz or up to THz) [45], [46]. Here, we estimated optical group index from the measured transmission spectrum of PhC WG shown in Fig. 2(f) (Fig. 5(a)) [47]. The group index of PhC WG is around 1.8, and increases dramatically near the band edge [48]–[50]. The group index in slow light region is comparable to RF index (Fig. 5(b)). For the frequency band of 60 GHz or lower, the RF index varies around 3.

B. Graphene Silicon p - i - n Modulator

For this hybrid structure, effective index variation of both graphene and silicon under forward bias is numerically characterized. For graphene, after adding the bias, the Fermi level of graphene is shifted, and carrier density is changed under bias.

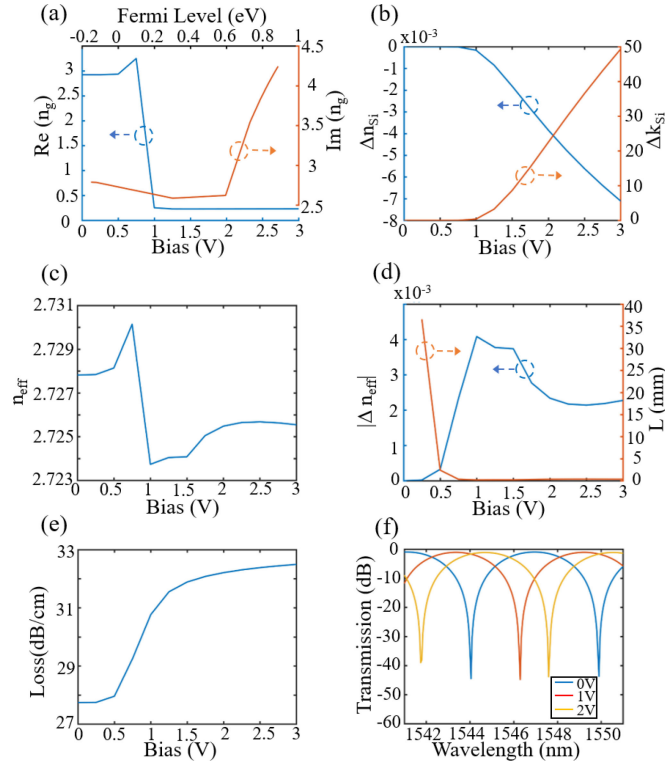


Fig. 6. G-Si *p-i-n* junction modulator. (a) Graphene Fermi level and refractive index variation under forward bias of 0 to 3V. (b) Real and imaginary part of silicon refractive index change under forward bias of 0 to 3V. (c) Effective index variation of hybrid structure under forward bias from 0 to 3V. (d) Total effective index change and the length to achieve π phase shift under forward bias. (e) Loss of the hybrid waveguide under forward bias from 0 to 3V. (f) Transmission spectrum of driving one arm of MZM around 1550 nm under bias 0 to 3V.

The changes in G Fermi level and complex refractive index are plotted in Fig. 6(a). For silicon under forward bias, the change of real and imaginary part of the silicon refractive index can be seen in Fig. 6(b). The total effective index of this hybrid structure is shown in Fig. 6(c). The total effective index change on the order of magnitude of 10^{-3} showing the good efficiency. And the corresponding length to achieve a π phase shift can be seen in Fig. 6(d). The minimum $V_{\pi} \cdot L$ is 0.133V·cm (2.67mm·0.5V) which shows this carrier rejection scheme of graphene integrated silicon modulator exhibiting very high efficiency in modulation.

For G-Si *p-i-n* modulator, the same methods are carried out for simulating carrier density changes and calculating the refractive index change for this hybrid structure. Under forward bias, the Fermi level of graphene is shifted and carrier density is changed. The calculated Fermi level, real and imaginary part of graphene refractive index are shown in Fig. 6(a). Under forward bias, real part of graphene refractive index increases then decreases as it experiences from interband to intraband transition [40], while the imaginary part increases. Due to the change of graphene conductivity, the real part of refractive index of graphene goes lower as the Fermi level increases. In silicon, the real part of silicon refractive index decreases while the imaginary part increases with forward bias (Fig. 6(b)). The change of total effective index of this hybrid structure is calculated on the

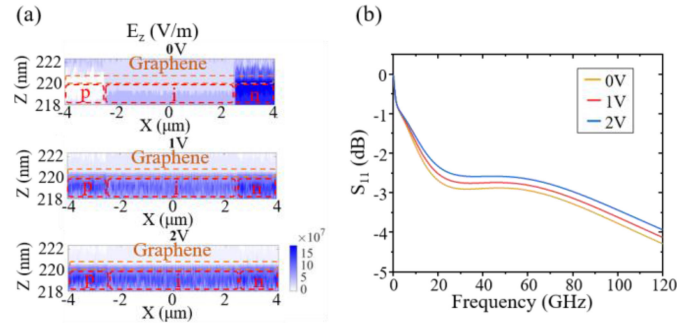


Fig. 7. (a) E_z component of built-in electric fields at graphene silicon interface under forward bias for G-Si *p-i-n* junction modulator. (b) Simulated S_{11} response G-Si *p-i-n* junction modulator under bias 0 to 2V.

10^{-3} scale with forward bias (Fig. 6(c)). Numerical calculation estimated that silicon dominates and contributes most 99.63% to the total effective index change comparing with graphene (0.27%). At low bias voltage ($<1V$), G contributes obviously to the total effective index change, while Si takes over the role beyond 1V. The $V_{\pi} \cdot L$ of the hybrid device is derived to be 1.19V·mm at 0.5V (2.397mm·0.5V) (Fig. 6(d)). This is due to significant variation of carrier densities in directly contacted silicon and graphene under bias. Along with the phase modulation, ~ 4 dB/cm incremental loss is observed at the reverse bias is set beyond 1 V (Fig. 6(e)). Assume the length of electro-optical interaction of each arm is 250 μ m, the transmission spectra of MZM with G-Si *p-i-n* junction phase modulators are calculated with different forward bias (Fig. 6(f)). Spectrum shift is about 1.5 nm per volt.

Simulated Built-in electric field component E_z can be seen in Fig. 7(a). By using the same method as G-Si *p-n* junction, the overall 3 dB bandwidth of EO response can be simulated. RC constant limited bandwidth can be obtained (about 80 GHz at 0V) from simulated S_{11} in Fig. 7(b). The carrier transit time is about 5 ps. The estimated EO bandwidth (limited by RC constant) is about 59 GHz.

In summary, directly contacted graphene silicon modulator has exhibited efficient and high-speed potential for EO modulator applications. By design of the electrical doping profile and co-optical design of graphene integrated photonic crystal waveguide, the hybrid device is able to achieve bandwidth of EO response at least 59 GHz and a very low $V_{\pi} \cdot L$ of 1.19 V·mm while suffering from considerable loss of over 25 dB/cm. This can be compromised by designing the proper length of MZM arm for electrical-optical interaction to achieve desired phase shift without too much loss.

C. Transparent Graphene Electrodes for Lithium Niobate Modulator

LN offers excellent electro-optic, acousto-optic, and non-linear optical properties for integrated active photonic devices [51]–[53]. Using crystal ion slicing technology, free-standing LN film with areas of several cm^2 were successfully fabricated on isolator layer and form lithium-niobate-on-isolator (LNOI) [4], [54], [55]. By fabricating a waveguide of dielectric on LN

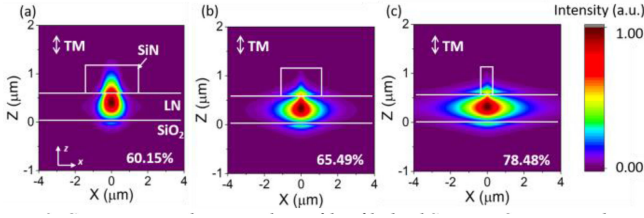


Fig. 8. Cross-sectional TM mode profile of hybrid SiN-LNOI waveguide with SiN width of (a) 2 μm , (b) 1 μm , and (c) 200 nm. The confinement factor in LN increases with decreased SiN width (a) 60.15%, (b) 65.49%, (c) 78.48%, respectively.

thin film, a single mode ridge waveguide can be formed on the continuous thin film of LN without etching the LN layer [56]. This approach has been demonstrated by using a handful of materials, such as silicon, tantalum pentoxide, titanium dioxide, chalcogenide glass (ChG), and silicon nitride (SiN) [56]–[58]. Compared to Si, the refractive index of silicon nitride (~ 1.9) is close to that of LN (~ 2.14 for extraordinary polarization at C band). This increases the confinement factor in the LN core and is beneficial for nonlinear interactions [59]. SiN has low linear absorption in visible-infrared transparency window and minimal two-photon absorption loss for incident photon in telecommunication bandwidth [60]. Since LN is anisotropic, the crystal orientation, optical polarization and electrode arrangement need to be co-designed for reaching low V_{π} . LN wafers are commercially available with many different orientations. Among them, X and Z-cut LNOI wafers are mostly common-used for integrated photonic applications [61]. The electro-optic coefficient γ_{33} (~ 31 pm/V) along Z-direction is larger than the other directions [62].

Since the LN is the active layer for the hybrid electro-optic modulator, a larger confinement factor in the LN is the primary goal of the optical design for maximizing the electro-optic tuning efficiency. Here we choose a 600-nm thickness LN thin-film on insulator substrate. The numerical simulation (performed with Lumerical MODE Solution) results show that the modal distribution or the confinement factor is much more sensitive to the SiN width than thickness (> 500 nm). Figure 8 shows the mode profile for the SiN waveguide width of 2 μm , 1 μm , and 0.2 μm , respectively. The correspondent confinement factor in LN layer is marked in the figures to be 60.15%, 65.49% and 78.48% respectively. Although the LN confinement factor increases with shrinking waveguide width, a hybrid structure with SiN width less than 0.2 μm does not support any guided mode, and thus we choose the structure in Fig. 8(c) as the optimized optical design as the starting point for the following electrical design.

Based on the optical design with maximum confinement factor in LN layer, vertically arranged (type I) and laterally arranged electrodes (type II) can be configured for Z- and X-cut LNOI modulator respectively, so the electrical field can be applied on the Z-direction with the maximum electro-optic coefficient.

The electrical design and simulated electrical field of the type I modulator are shown in Fig. 9(a). The simulations were carried out in COMSOL RF module, where the SiN is 800 nm thick and 200 nm wide. The graphene layer is sandwiched between SiN

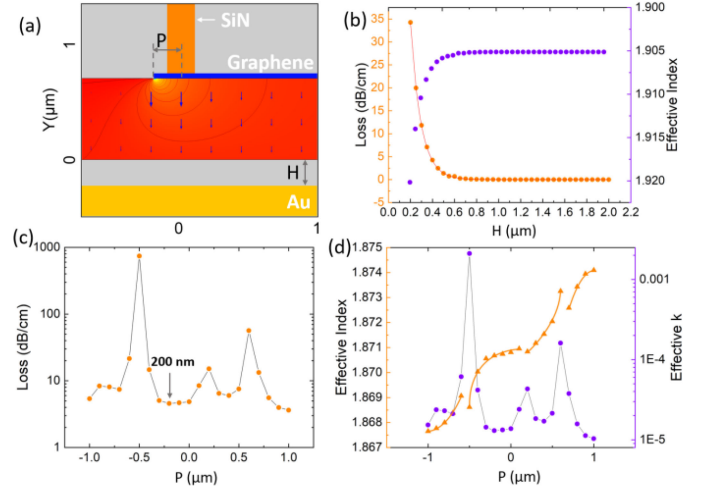


Fig. 9. Device geometry dependent propagation loss with graphene between SiN and LN layer. (a) Simulated electrical field on the cross-section. The position (b) Propagation loss versus oxide thickness (H), (c) graphene edge-waveguide center distance (P). (d) Real (left) and imaginary (right) part of the effective index versus P.

and LN, and the other electrode is the metal layer under 600 nm silicon oxide. There are two key parameters, oxide thickness and graphene edge position (marked as H and P in Fig. 9(a)) in the design, which needs to be carefully selected for efficient electro-optic tuning without sacrificing optical propagation loss.

Fig. 9(b) shows the propagation loss versus the oxide thickness (H). H is set from 0.2 μm to 2 μm (orange dots). The data was fitted as an exponential function (orange curve). The fitting result shows light absorption of the Au electrode is negligible as H is larger than 600 nm, resulting 1.905 effective index in the hybrid waveguide (purple dots in Fig. 9(b)).

Another key parameter is the relative position of waveguides and graphene edge (P). Plasma resonance will occur and induce coupling effect between guiding modes in LN and plasmonic modes in graphene, which cause additional loss (Fig. 9(c)). It can be seen that the lowest loss (~ 4.58 dB/cm) appears at $P = 200$ nm. Fig. 9(d) shows the correspondent real (orange dots) and imaginary (violet dots) part of the effective index for the hybrid waveguide. The higher propagation loss is observed as graphene edge placed ~ 500 nm away from the waveguide center, which might be attributed to the excitation of plasmonic mode in graphene [63]. The plasmonic loss can be minimized by placing the graphene electrode at positions from 0 to 300 nm.

Type II electrode design is illustrated in Fig. 10(a). The horizontally arranged electrodes are more commonly used in integrated devices as it avoids complexity of embedding metal layer under oxide. Voltage is applied between the two separated graphene electrodes. Same as type I, the presence of graphene layer leads to additional loss. Fig. 10(b) shows the optical propagation loss versus graphene electrode intervals between 0 to 2 μm .

In two types of designs, graphene can be retreated as an equipotential surface of electrodes, and its refractive index change contributes little to the total effective index. Figs 11(a) and 11(b) show integral electro-optic overlap factors of types

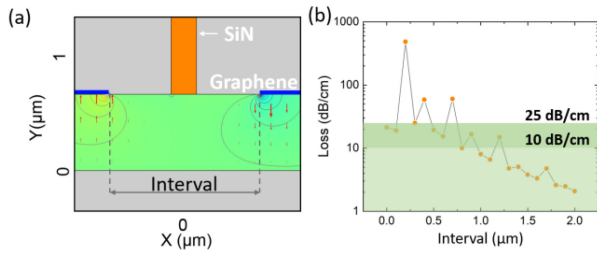


Fig. 10. (a) Simulated electrical field profile in LN layer for type II design. (b) Optical loss versus graphene electrodes interval.

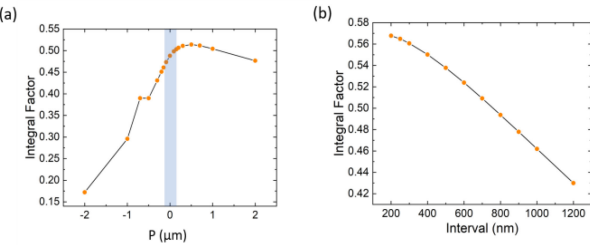


Fig. 11. (a) Electro-optic integral factor versus P for type I design. (b) The integral factor versus interval for type II design.

I and II designs versus device design parameters. Maximum integral factor of 0.52 ($P = 200$ nm) and 0.57 (Interval = 200 nm) are achieved in type I and II design respectively (orange dots). Compared to lateral metal-LN waveguide-metal designs (integral factor is estimated to be 20%), the integral factor is 2.5 times better, and thus the predicted $V_{\pi} \cdot L$ should be dropped to 1/4 in our designs.

III. CONCLUSION AND DISCUSSION

Here we proposed and compared a few device architectures for energy-efficient and ultrafast integrated phase modulators of G. With active Si substrate, G refractive index modulation enables the fast and efficient charge carrier transport for the high speed and efficiency optoelectronic devices comparing with monolithic material platforms. Specifically, experimental demonstrations show more than 40 GHz bandwidth in G-Si *p-i-n* structure. And it could be improved further by optimizing the doping profiles in Si and graphene covering position. A phase modulator with optoelectronic bandwidth >60 GHz and low $V_{\pi} \cdot L$ (0.119V·cm) can be simultaneously achieved. On a dielectric material platform, TM polarized mode launched in SiN-LN structure can minimize the linear absorption in G, with maximized EO modulation efficiency in Z-cut LN. The proposed device structure avoids direct etching in LN while maintaining ~70% optical field confinement in LN region. Compared with monolithic LN modulators, both lateral and vertical electrodes design with G show 2.5 times reduction of $V_{\pi} \cdot L$.

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